

■ Array

`Array[f, n]` generates a list of length n , with elements $f[i]$.

`Array[f, { $n_1$ ,  $n_2$ , ...}]` generates a $n_1 \times n_2 \times \dots$ array of nested lists, with elements $f[i_1, i_2, \dots]$.

`Array[f, dims, origin]` generates a list using the specified index origin (default 1).

`Array[f, dims, origin, h]` uses head h , rather than `List`, for each level of the array.

Example: `Array[f, 3]  $\longrightarrow$  {f[1], f[2], f[3]}`.

■ `Array[f, {2, 3}]  $\longrightarrow$  {{f[1, 1], f[1, 2], f[1, 3]}, {f[2, 1], f[2, 2], f[2, 3]}}` generates a 2×3 matrix. ■ `Array[#1^#2 &, {2, 2}]  $\longrightarrow$  {{1, 1}, {2, 4}}`. ■ `Array[f, 3, 0]  $\longrightarrow$  {f[0], f[1], f[2]}` generates an array with index origin 0. ■ `Array[f, 3, 1, Plus]  $\longrightarrow$  f[1] + f[2] + f[3]`. ■ Note that the dimensions given to `Array` are *not* in standard *Mathematica* iterator notation. ■ See page 182. ■ See also: `Table`.